Game Theory Meets Computer Science

Part Three Auction & Computational Social Choice

姜桂飞
南开大学 · 软件学院
G.Jiang@nankai.edu.cn

Motivation: Algorithmic Mechanism Design

Something goes wrong?

you \ pair	α	β
α	(0, 0)	(3, -1)
β	(-1, 3)	(1, 1)

- Dominant-strategy equilibrium: (α,α) → (B-,B-) Pareto inefficient!!
- Games like this called Prisoners' Dilemmas.

The Battle of the Sexes

Girl \ Boy		
	(2, 1)	(0, 0)
	(0, 0)	(1, 2)

- Two NEs (夺冠, 夺冠), (八佰 ,八佰)
- Which one to choose?

- Design games that have both good game-theoretical and algorithmic properties.

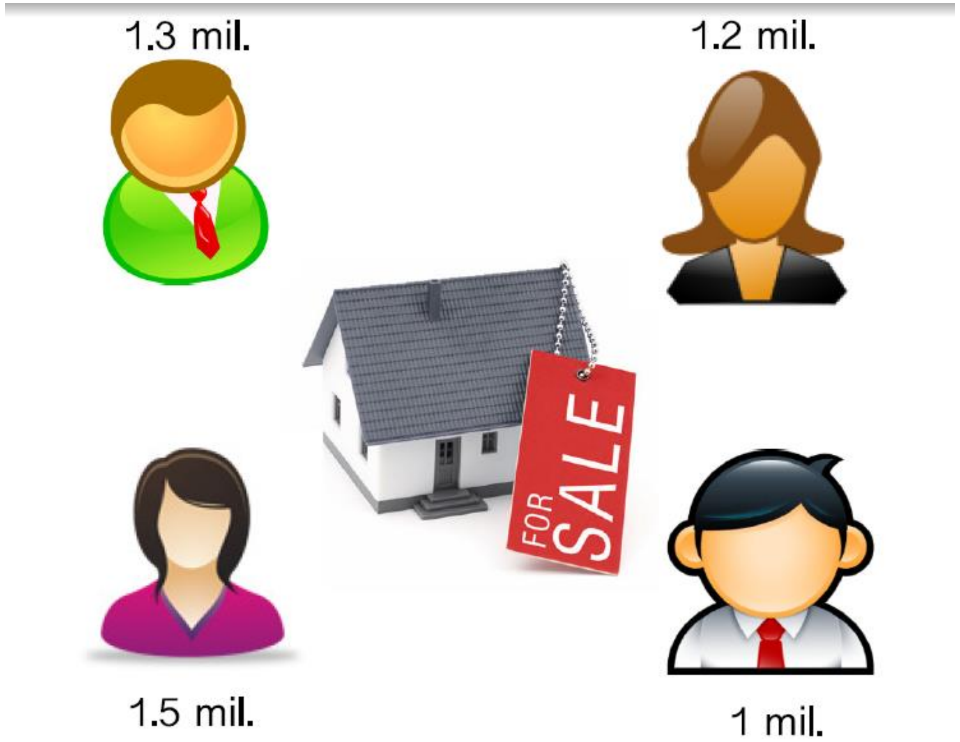
Let’s design a game...

Design Goal

How can a house-seller sell her house with the "highest" profit?

A Game Design Example

- Design Goal: How can a house-seller sell her house with the "highest" profit?



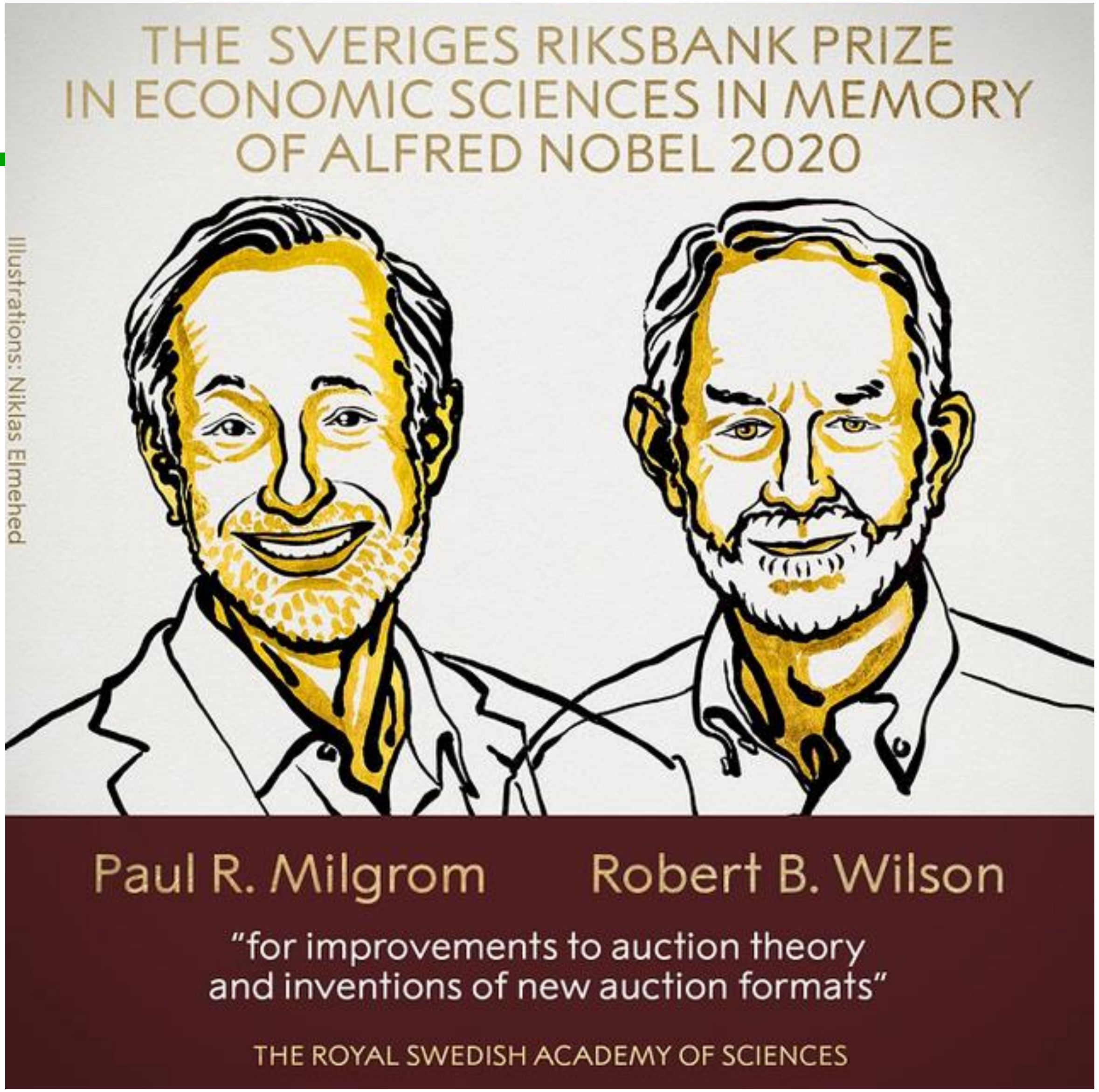
- Challenge: the seller doesn't know how much the buyers are willing to pay (their valuations).

A Game Design Example

- Design Goal: How can a house-seller sell her house with the "highest" profit?

Auction

- An auction takes place between an agent known as the auctioneer and a collection of agents known as the bidders.
  - Each bidder i has a nonnegative private valuation v_i
  - If a bidder i loses an auction, her utility is 0. If the bidder wins at a price p, her utility is $v_i - p$.
- The goal of the auction is for the auctioneer to allocate the good to one of the bidders.
- In most settings the auctioneer desires to maximize the price; bidders desire to minimize price.



威廉·维克里（William Vickrey，1914年6月21日—1996年10月11日）1996年诺贝尔经济学奖获得者

Typical Auction Protocols

- English Auction
- Dutch Auction
- First-price sealed-bid auction
- Vickrey auction

A Game Design Example

- Design Goal: How can a house-seller sell her house with the "highest" profit?

Solution: Second Price Auction
(Vickrey Auction/VCG)

- Input: each buyer reports a price/bid to the seller
- Output: the seller decides
  - allocation: the agent with the highest report price wins.
  - payment: the winner pays the second highest price reported.

Ideal Auctions

Ideal Auctions

- Truthful: buyers will report their highest willing payments
- Social Welfare Maximization： the mechanism makes the sum of the utilities of each agent to be maximal.
- Pareto efficiency: an outcome is Pareto efficient if there is no other outcome that will make at least one agent better off without making at least one other agent worse off.
- Computational Efficiency: time polynomial (indeed, linear)

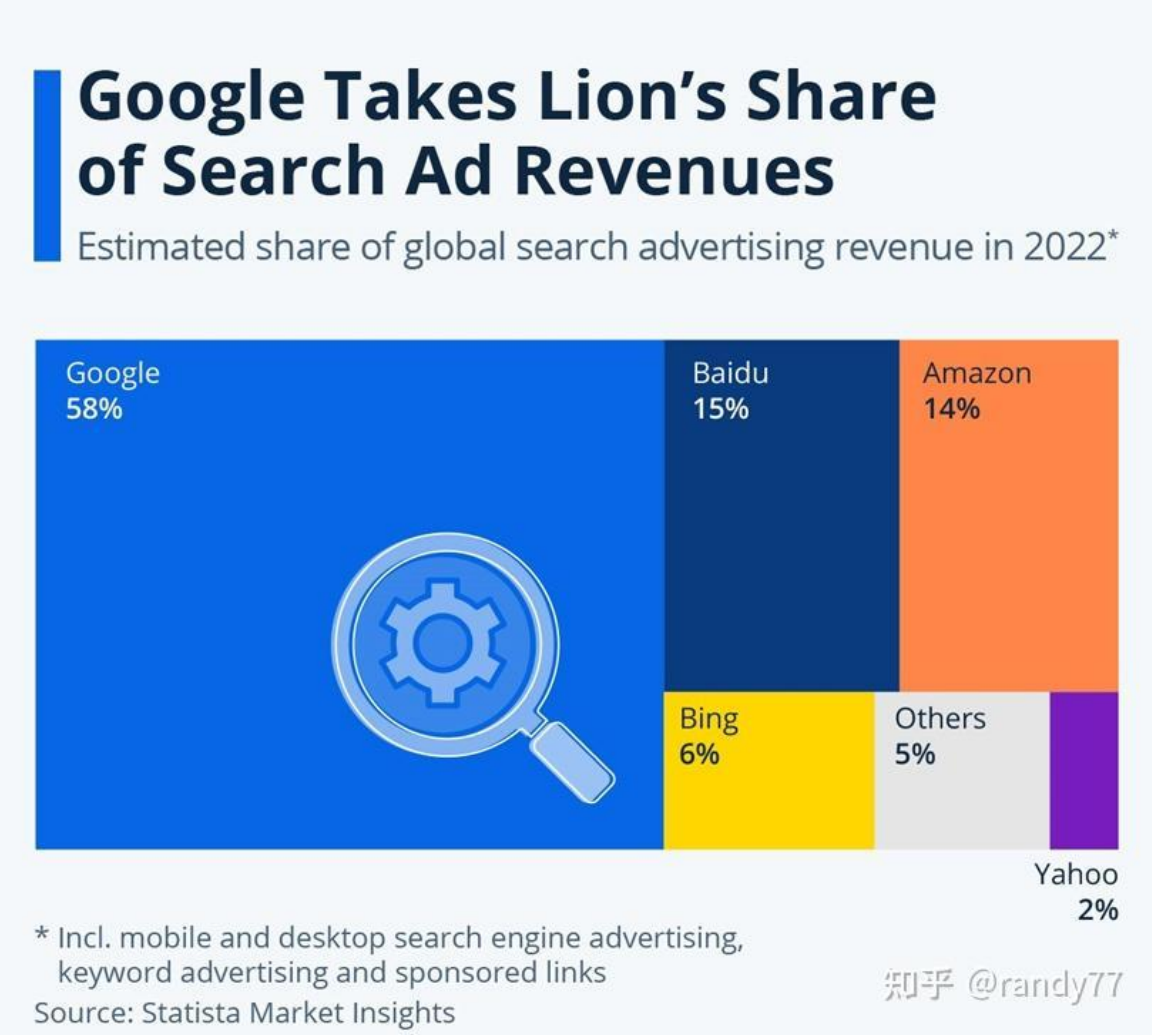
Generalized Second Price (GSP) Auction

How does the ad auction work?

- An advertiser chooses a set of keywords that are related to the product it wishes to sell.
- Each advertiser states a bid for each keyword that can be interpreted as the amount that it is willing to pay if a user clicks on its ad.
- When a user's search query matches a keyword, a set of ads is displayed. These ads are ranked by bids and the ad with the highest bid receives the best position; i.e., the position that is mostly likely to be clicked on by the user.
- If the user clicks on an ad, the advertiser is charged an amount that depends on the bid of the advertiser below it in the ranking.

Ad auctions: second-price

- Introduced by Google in 2002.
- Every advertiser submits a bid.
- Advertisers are arranged on the page in descending order of their bids.
- The advertiser in the first position pays a price per click that equals the bid of the second advertiser plus an increment (say 1¢); the second advertiser pays the price offered by the third advertiser plus an increment and so forth
- This auction mechanism is named as Generalized Second-Price Auction (GSP auction).



谷歌2024年Q1季报

广告收入：Q1收入为617亿元，同比增长13%，连续2个季度增长超过10%，广告收入占比76.6%，其中：

搜索广告：Q1收入为462亿元，同比增长14.2%；

Youtube广告：Q1收入为81亿元，同比增长20.9%

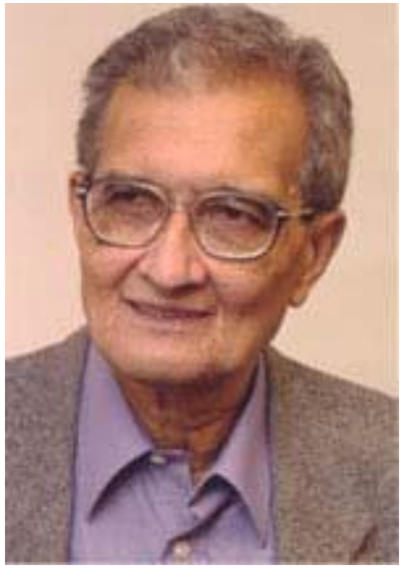
Computational Social Choice

What is Social Choice Theory

- Social choice theory： Mathematical theory for aggregating in dividual preferences into collective decisions.

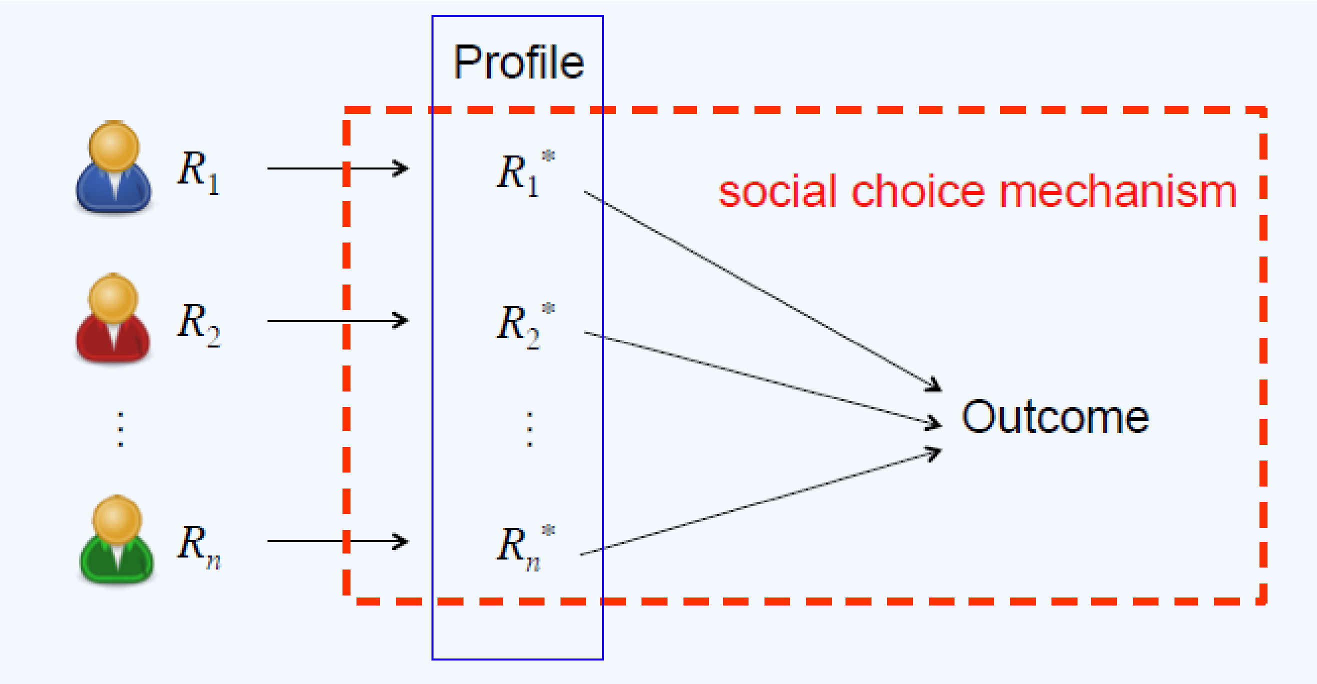
Social Choice Theory

- Originated in ancient Greece
- Formal foundations
- 18th Century (Condorcet and Borda)
- 19th Century: Charles Dodgson (a.k.a. Lewis Carroll)
- 20th Century: Nobel prizes to Arrow (1972) and Sen(1998)



Social Choice Theory

- Want to select a collective outcome based on (possibly different) individual preferences
  - Presidential election
  - restaurant/movie selection for group activity
  - committee selection
  - facility location, …



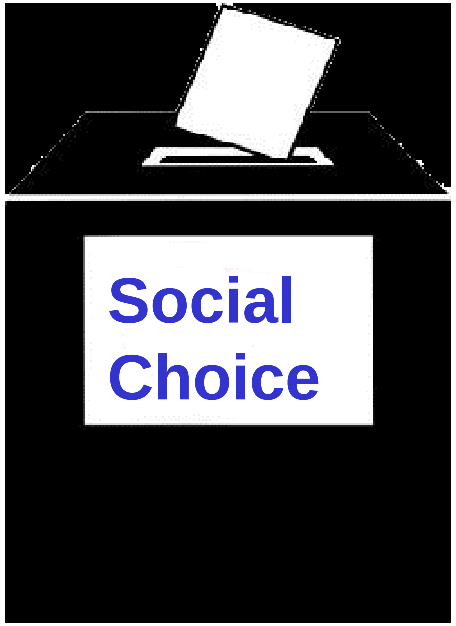
Computational social choice (COMSOC)

- “*Computational social choice is an interdisciplinary field of study at the interface of social choice theory and computer science, promoting an exchange of ideas in both directions.*”

---http://www.illc.uva.nl/COMSOC/

Social Choice and Computer Science

Computational thinking + optimization algorithms



Strategic thinking + methods/principles of aggregation

- Computational social choice emphasizes the fact that any method of decision making is ultimately an algorithm.

IJCAI 2024 Computers and Thought Award

- **Nisarg Shah**, Associate Professor of Computer Science, University of Toronto,
- For his contributions to AI and society, in particular foundational work on the theory of algorithmic fairness using principles from social choice theory.

Three Voting Rules

- Voting is the prototypical form of collective decision making
- Plurality: elect the candidate ranked first most often (i.e., each voter assigns one point to a candidate of her choice, and the candidate receiving the most votes wins)
- Borda: each voter gives $m-1$ points to the candidate she ranks first, $m-2$ to the candidate she ranks second, etc., and the candidate with the most points wins
- Approval: voters can approve of as many candidates as they wish, and the candidate with the most approvals wins

Borda count in real life

Current uses [edit]

Political uses [edit]

The Borda count is used for certain political elections in at least three countries, Slovenia and the tiny Micronesian nations of Kiribati and Nauru. In Slovenia, the Borda count is used to elect two of the ninety members of the National Assembly: one member represents a constituency of ethnic Italians, the other a constituency of the Hungarian minority. As noted above, members of the Parliament of Nauru are elected based on a variant of the Borda count that involves two departures from the normal practice: (1) multi-seat constituencies, of either two or four seats, and (2) a point-allocation formula that involves increasingly small fractions of points for each ranking, rather than whole points. In Kiribati, the president (or *Beretitenti*) is elected by the plurality system, but a variant of the Borda count is used to select either three or four candidates to stand in the election. The constituency consists of members of the legislature (*Maneaba*). Voters in the legislature rank only four candidates, with all other candidates receiving zero points. Since at least 1991, tactical voting has been an important feature of the nominating process.

The Republic of Nauru became independent from Australia in 1968. Before independence, and for three years afterwards, Nauru used instant-runoff voting, importing the system from Australia, but since 1971, a variant of the Borda count has been used.

The modified Borda count has been used by the Green Party of Ireland to elect its chairperson.[6][7]

The Borda count has been used for non-governmental purposes at certain peace conferences in Northern Ireland, where it has been used to help achieve consensus between participants including members of Sinn Féin, the Ulster Unionists, and the political wing of the UDA.

Other uses [edit]

The Borda count is used in elections by some educational institutions in the United States.

- University of Michigan
  - Central Student Government
  - Student Government of the College of Literature, Science and the Arts (LSASG)
- University of Missouri: officers of the Graduate-Professional Council
- University of California Los Angeles: officers of the Graduate Student Association
- Harvard University: officers of the Civil Liberties Union
- Southern Illinois University at Carbondale: officers of the Faculty Senate,
- Arizona State University: officers of the Department of Mathematics and Statistics assembly.
- Wheaton College, Massachusetts: faculty members of committees.
- College of William and Mary: members of the faculty personnel committee of the School of Business Administration (tie-breaker).

The Borda count is used in elections by some professional and technical societies.

- International Society for Cryobiology: Board of Governors.
- Tempo sustainable design network: management committee.
- U.S. Wheat and Barley Scab Initiative: members of Research Area Committees.
- X.Org Foundation: Board of Directors.

The OpenGL Architecture Review Board uses the Borda count as one of the feature-selection methods.

The Borda count is used to determine winners for Toastmasters International speech contests. Judges offer a ranking of their top three speakers, awarding them three points, two points, and one point, respectively. All unranked candidates receive zero points.

The modified Borda count is used to elect the President for the United States member committee of AIESEC.

The Borda count, and points-based systems similar to it, are often used to determine awards in competitions.

The Borda count is a popular method for granting sports awards in the United States. Uses include:

- MLB Most Valuable Player Award (baseball)
- Heisman Trophy (college football)[8]
- Ranking of NCAA college teams

The Eurovision Song Contest uses a positional voting method similar to the Borda count, with a different distribution of points: only the top ten entries are considered in each ballot, the favorite entry receiving 12 points, the second-placed entry receiving 10 points, and the other eight entries getting points from 8 to 1. Although designed to favor a clear winner, it has produced very close races and even a tie.

The People's Remix Competition uses a Borda variant where each voter ranks only the top three contestants.

The Borda count is used for wine trophy judging by the Australian Society of Viticulture and Oenology, and by the RoboCup autonomous robot soccer competition at the Center for Computing Technologies, in the University of Bremen in Germany.

The Finnish Associations Act lists three different modifications of the Borda count for holding a proportional election. All the modifications use fractions, as in Nauru. A Finnish association may choose to use other methods of election, as well.[9]

Voting

n voters…

… each produce a ranking of *m* alternatives…

$b \succ a \succ c$

$a \succ c \succ b$

$a \succ b \succ c$

… which a social preference function (SPF) maps to one or more aggregate rankings.

$a \succ b \succ c$

… or, a social choice function (SCF) just produces one or more winners.

a

Plurality

1 0 0

$b \succ a \succ c$

$a \succ c \succ b$

$a \succ b \succ c$

$a \succ b \succ c$

2 1 0

Borda

$2 \quad 1 \quad 0$

$b \succ a \succ c$

$a \succ b \succ c$

$a \succ c \succ b$

$5 \quad 3 \quad 1$

$a \succ b \succ c$

Choosing a rule

- How do we choose a rule from all of these rules?
- How do we know that there does not exist another, “perfect” rule?
- Let us look at some criteria that we would like our voting rule to satisfy

Condorcet criterion

- A candidate is the Condorcet winner if it wins all of its pairwise elections
- Does not always exist…
- … but the Condorcet criterion says that if it does exist, it should win

- Many rules do not satisfy this
- E.g., for plurality:
  - b > a > c > d
  - c > a > b > d
  - d > a > b > c
- a is the Condorcet winner, but it does not win under plurality

Majority criterion

- If a candidate is ranked first by a majority (> ½) of the votes, that candidate should win
  - Relationship to Condorcet criterion?

- Some rules do not even satisfy this
- E.g., Borda:
  - a > b > c > d > e
  - a > b > c > d > e
  - c > b > d > e > a
- a is the majority winner, but it does not win under Borda

Independence of irrelevant alternatives

- Independence of irrelevant alternatives criterion: if
  - the rule ranks a above b for the current votes,
  - we then change the votes but do not change which is ahead between a and b in each vote

  then a should still be ranked ahead of b.
- None of our rules satisfy this

Arrow’s impossibility theorem [1951]

- Suppose there are at least 3 candidates
- Then there exists no rule that is simultaneously:
  - Pareto efficient (if all votes rank a above b, then the rule ranks a above b),
  - nondictatorial (there does not exist a voter such that the rule simply always copies that voter’s ranking), and
  - independent of irrelevant alternatives

Manipulability

- Sometimes, a voter is better off revealing her preferences insincerely, AKA manipulating
- E.g., plurality
  - Suppose a voter prefers a > b > c
  - Also suppose she knows that the other votes are
    - 2 times b > c > a
    - 2 times c > a > b
  - Voting truthfully will lead to a tie between b and c
  - She would be better off voting, e.g., b > a > c, guaranteeing b wins

Judgment aggregation

- Three judges have to decide on a case of an alleged breach of contract
- They need to decide (a) whether the contract is **valid** and (b) whether the contract has been **breached**.
- Legal doctrine stipulates that the defendant is **liable** if and only if (a) and (b) hold.

	Valid?	*Breach?*	*Liable?*
Judge 1	**Yes**	**Yes**	**Yes**
Judge 2	**Yes**	**No**	**No**
Judge 3	**No**	**Yes**	**No**

Why is this considered a paradox?

	p	q	$p \wedge q$
Judge 1	**Yes**	**Yes**	**Yes**
Judge 2	**Yes**	**No**	**No**
Judge 3	**No**	**Yes**	**No**
Majority	**Yes**	**Yes**	**No**

- **Reason 1: Premise-based procedure** and **conclusion-based procedure** produce different outcomes.
- **Reason 2:** Even though each individual judgment is logically consistent, the **majority outcome** is not.

How to design a good social choice mechanism?

WHAT IS BEING "GOOD"?

Two goals for social choice mechanisms

GOAL1: democracy

GOAL2: truth

Challenges

- Goodness:
  - democracy: fairness, efficiency, etc
  - truth: accuracy
- Computation: how can we compute the outcome as fast as possible
- Incentives: what if an agent does not report her true preferences?

Computational Social Choice and the Web

robovote.org

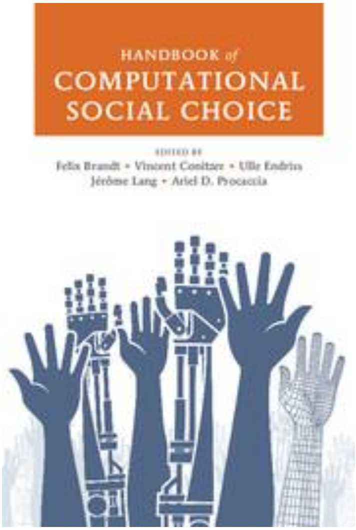
spliddit.org

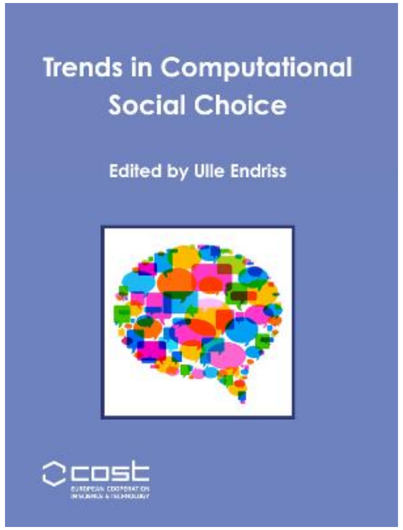
whale.imag.fr

Building tools to allow users to interact directly with social choice algorithms on the web is not only useful for those users but also an opportunity to collect data and get ideas for new research questions.

Finding out about New Developments

- The Handbook of COMSOC (2016) represents the state of the art around 2012, when it was conceived.
- Trends in COMSOC (2017) covers several important developments that have taken place since then.

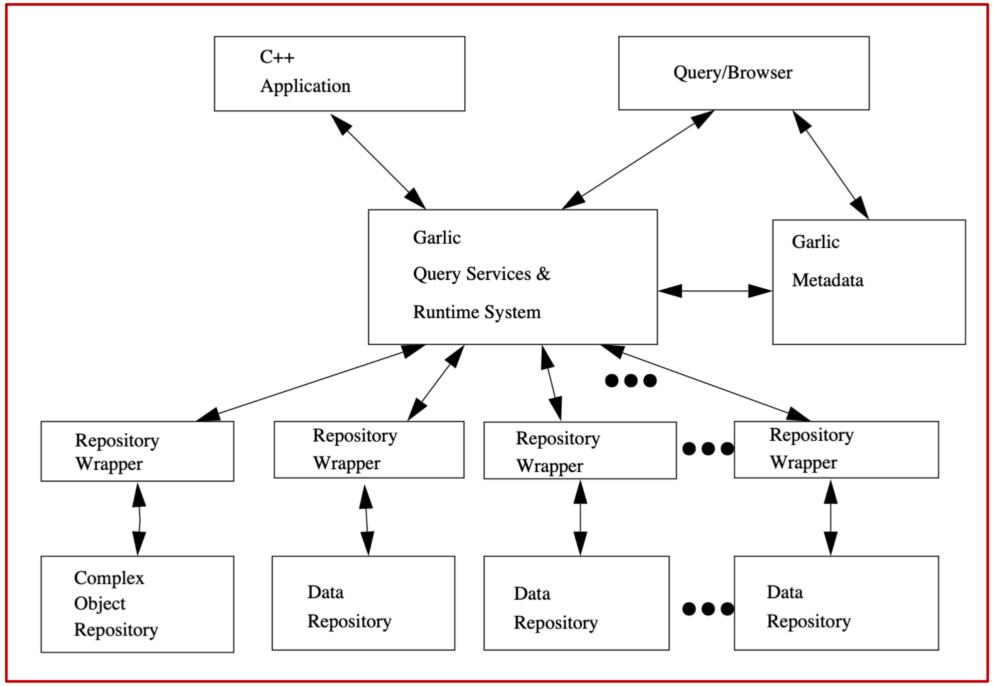




- A lot of work in COMSOC gets published at major AI conferences: AAMAS is the most important multiagent systems conference IJCAI, AAAI, ECAI are the main general-purpose AI conferences
- At the interface with Algorithmic Game Theory (and Theoretical Computer Science more generally), the most important conference is EC.
- In Computer Science most new ideas (first) show up at conferences, but also look at the corresponding journals (JAIR, AIJ, TEAC, JAAMAS).

Research Topics

- Aggregation mechanism design in social networks
  - Formalization and automation
  - Database Information Combination
- Anti-Manipulation
  - Complexity
- Beyond
  - Recommendation Systems
  - Ontology Merging



Publications: AAMAS-2015, PRIMA-2014, Journal of Computational Intelligence 2018, AAAI-2020, IJCAI-2020, ADMA-2023